

# Megha Patel

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## EDUCATION

**University of the Pacific** | Master's in Computer Science | **GPA: 3.94/4.0** Stockton, California  
Courses: Software Engineering, Web Development, Machine Learning, Data Science, UI/UX Design 2022 – 2024  
**Gujarat Technological University** | Bachelor's degree in Computer Engineering | **GPA: 8.81/10** Gandhinagar, Gujarat  
Courses: Data Structures and Algorithms, OOP, DBMS, OS, ML/DL, NLP, Cloud Computing 2018 – 2022

## WORK EXPERIENCE

**Founding Software Engineer @ NJT Automation** Apr 2025–Present

- Led development of a scalable **repair marketplace platform** using **React**, **TypeScript**, and **Supabase**, implementing modular **role-based access** and designing **normalized PostgreSQL** schemas to support multi-role user flows.
- Implemented real-time messaging system using **WebSocket connections** with data validation and anomaly detection, achieving **sub-200ms message delivery** with **99.8% reliability** across concurrent user sessions.
- Designed **event-driven notification infrastructure** leveraging **PostgreSQL triggers** and **email service integrations** to deliver **automated status updates** throughout multi-step repair workflows.
- Established **CI/CD pipelines** with **GitHub Actions** and **Docker**, decreasing deployment times by **40%** and improving build consistency across multiple environments.
- Optimized** application performance by **35%** through **advanced React Query caching**, **code splitting**, and **lazy loading**, optimizing bundle size, reducing page load times and improving user experience.

**Full Stack Developer @ Reel Talk** Aug 2024–Apr 2025

- Architected **RESTful API infrastructure** using **Node.js** and **TypeScript**, processing **1,000+** daily requests to serve TMDB-powered movie data across trending communities, user forums, and personalized watchlists.
- Designed microservices** for modular handling of **movie metadata**, **user management**, and **discussion forums**, improving modularity and enabling independent scaling across services.
- Built and integrated** a personalized Watchlist feature in the frontend using React and TypeScript, allowing users to create, update, and share collections of movies/TV shows with **seamless TMDB sync**, enhancing user engagement.
- Wrote unit and integration tests** for backend APIs using **Jest** and **Supertest**, achieving **85% coverage** and reducing post-deployment bugs by **30%**.

**Research and Development Intern @ University of the Pacific** Oct 2023–May 2024

- Managed cross-functional team** of 5 developers and analysts to deliver a public-facing data visualization platform for **San Joaquin County** and its 7 cities, commissioned by the San Joaquin Council of Governments (**SJCOG**).
- Delivered key features using **Angular**, integrating **Highcharts** for interactive charts and **FRED/Census APIs** for real-time ingestion of **50+ socio-economic indicators** with automated data validation.
- Conducted stakeholder demos and **user testing sessions**, iterating on UX/UI based on feedback and contributing to a **200% increase in user engagement** and broader **regional adoption** post-launch.

**Software Engineer Intern @ BrainyBeam Technologies** Jan 2022–May 2022

- Developed a **sales forecasting** application using **Python**, **Django**, and **Scikit-learn** with **time-series modeling**, implementing **linear regression** algorithms that improved quarterly revenue prediction accuracy by **15%**.
- Leveraged **AWS EC2** and **RDS** infrastructure to optimize sales forecasting model deployment and data processing, reducing computation time by **35%** for large historical datasets and enabling **real-time prediction** capabilities.
- Collaborated** with senior engineers on **code reviews** and **agile development** practices, contributing to 3 major feature releases while maintaining comprehensive unit tests using **Django's testing framework**.

## TECHNICAL SKILLS

**Programming Languages:** JavaScript, TypeScript, Python, Java, SQL  
**Technologies:** React.js, Next.js, Node.js, Express.js, Git, Docker, Firebase, GCP, AWS, MongoDB, MySQL, PostgreSQL, DynamoDB, Tailwind CSS, Bootstrap, GraphQL, FastAPI, Django, RESTful APIs, Numpy, Pandas  
**Core Skills:** API Design, System Design, Microservices Architecture, Testing & Debugging, Agile Methodologies, CI/CD, Performance Optimization, Machine Learning, LLM, Prompting

## PROJECTS

**LLM-Powered Therapy Training Platform** | *React, Flask, Firebase, OpenAI API, JWT*

- Built a virtual therapy simulator where therapists interact with simulated patients, featuring **experience-based UX routing**, **secure authentication** with **JWT**, and session tracking via **Firebase Firestore**.
- Integrated **OpenAI's API** for real-time, **persona-driven chat** streaming, implemented in-session message history, API failure handling, and a responsive frontend using React with custom chat UI components.

**Interactive PDF Chatbot for Extractive QA** | *Langchain, Gradio, PyPDF, Transformers, FAISS*

- Designed a **robust interactive PDF chatbot** leveraging Langchain and Gradio, implementing a **RAG** framework that boosted query resolution speed by **60%** and improved content retrieval **precision by 50%**.

**Academic Support Hub** | *MERN Stack*

- Engineered a centralized **student portal** that simplifies **discovery of TA** availability and course info, significantly improving student access to timely academic support.

## PUBLICATION

- Lightweight ML-enabled Intrusion Detection Framework for Vehicular Ad-Hoc Networks @ [IEEE AIC'24](#)